

**Time:** 5 minutes**Outcome:** Interpret LRAC, Economies & Diseconomies of Scale.**Difficulty:**

●●○ Intermediate

LRAC, Economies & Diseconomies of Scale



THE CORE IDEA

Long-run average cost (LRAC) shows the lowest attainable cost per unit when every input and plant size can vary. Scale can make specialized tasks and equipment economical, lowering LRAC over a range. Diminishing marginal product is a short-run concept; diseconomies concern scale when all inputs can vary. Falling long-run average cost as output expands is economies of scale.



HOW TO RECOGNIZE IT

- A downward LRAC segment means economies of scale; an upward segment means diseconomies.
- In the long run, every input and plant size can change.
- Diminishing marginal product instead holds at least one input fixed.

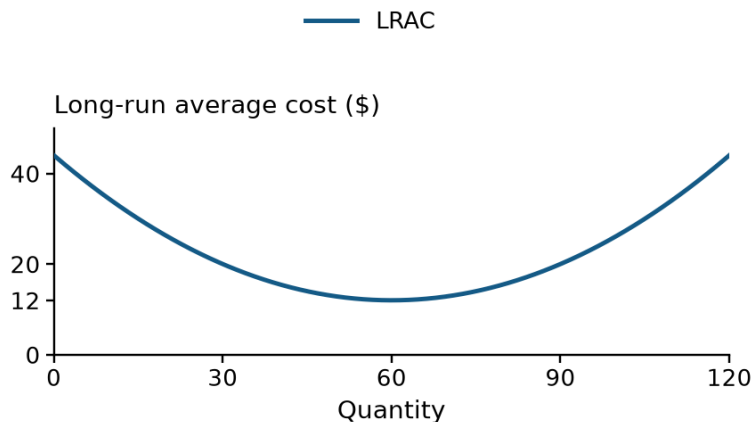


WATCH OUT

Do not confuse scale and diminishing returns. Diminishing marginal product is a short-run concept; diseconomies concern scale when all inputs can vary.



WORKED EXAMPLE: READING LONG-RUN AVERAGE COST



The LRAC curve falls from about \$20 at 30 units to its minimum of about \$12 at 60 units, showing economies of scale. It then rises back toward \$20 by 90 units, showing diseconomies of scale. The cost-minimizing scale is approximately 60 units.



CHECK YOURSELF

At $Q = 30$ the LRAC curve is falling, while at $Q = 90$ it is rising. Which point shows economies of scale and which shows diseconomies?

**READY?**

Return to the game and master this concept.